Fraud Risk & Transaction Analytics

Key finding: Fraud is only 0.52% of transactions but 3.95% of dollar volume, so fraud tickets are about 7-8x larger than normal ones. Late 2020 rate may be affected by data coverage. (All Data)

category
All

Year
All

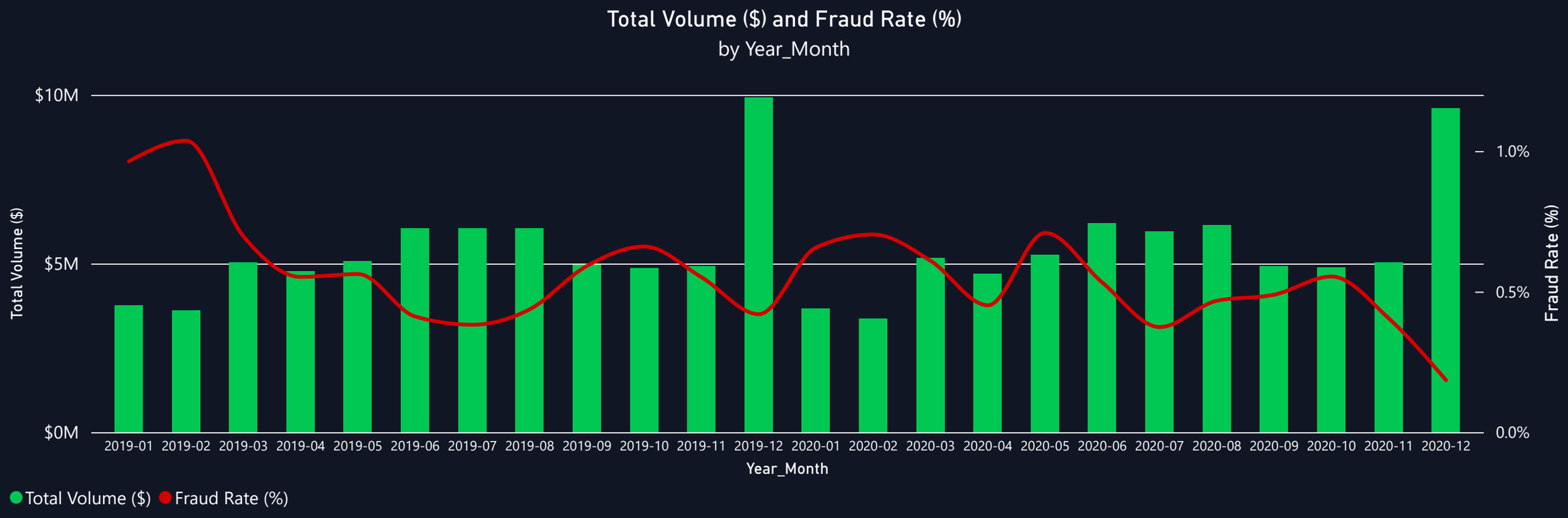
state
All

Total Volume ($)
$129.79M

Fraud Loss ($)
5.12M

Fraud Rate by Value (%)
3.95%

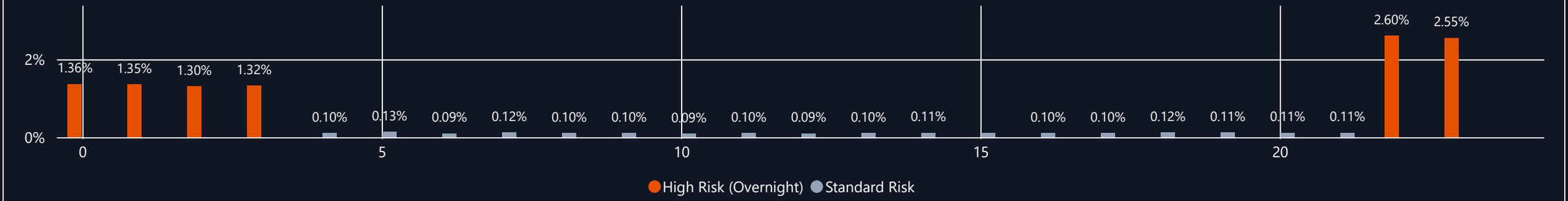
Fraud Rate by Count(%)
0.52%



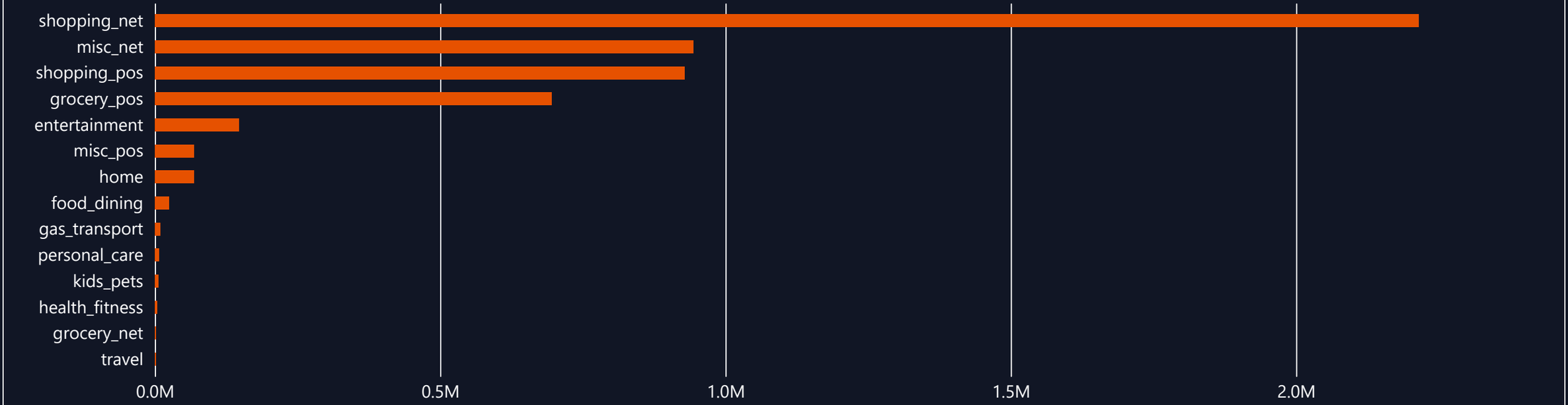
Key finding: Hours 22-03 are about 23% of transactions but about 85% of all fraud. Hours 22-23 run at 2.6%, five times the 0.52% average.

Hourly Fraud Rate (%)

by Hour and Time_Risk_Window



Fraud Loss ($) by Category



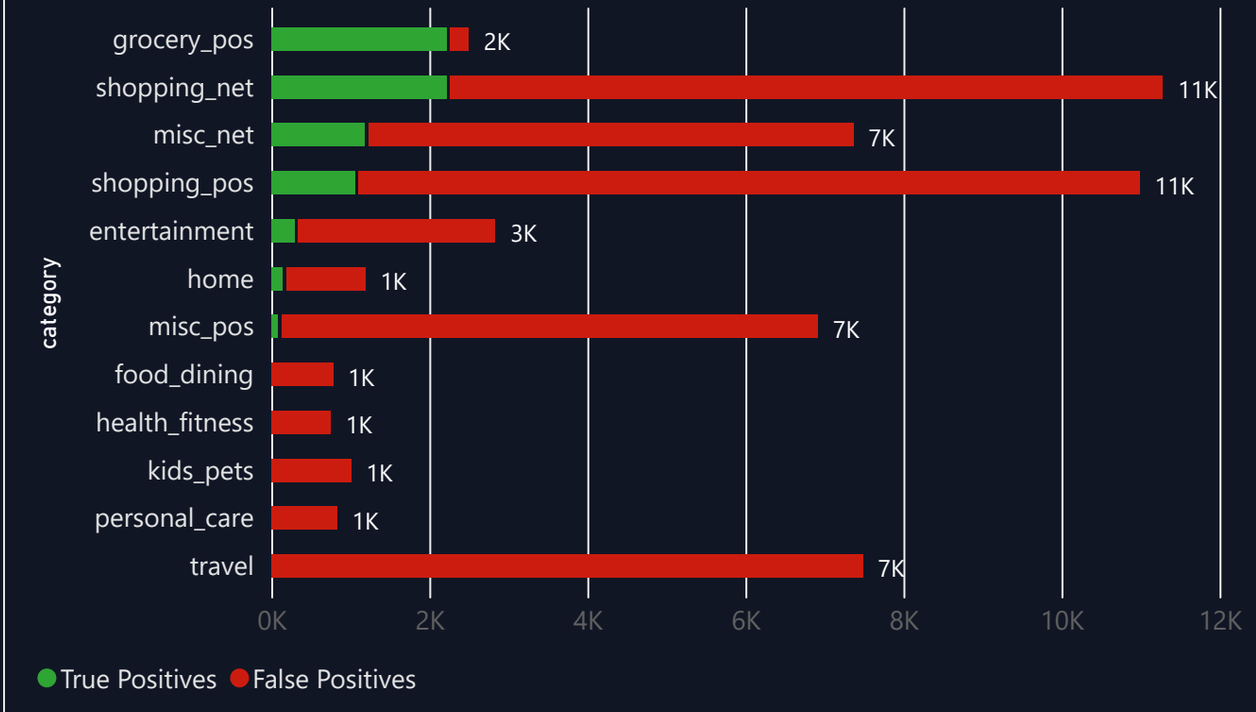
Rule Performance & Simulation Console

Rule Precision (%)	Rule Recall (%)	Rule F1 (%)	False Pos. Rate(%)
13.3%	74.5%	22.6%	2.5%

Threshold

250

True Positives and False Positives
by category



True Positives (fraud caught)

7K

False Positives (false alarms)

47K

False Negatives (fraud missed)

2K

True Negatives (correctly cleared)

1,796,012

Amount-only rule vs Combined Rule

Key finding (at $250): The combined rule cuts false alarms by about 92% and triples precision (13.3% to 42.3%), but catches 30% of fraud instead of 74.5%.

Threshold

250

Amount-only rule

Rule Precision (%)	Rule Recall (%)	Rule F1 (%)	False Positives
13.3%	74.5%	22.6%	46,731

Combined rule

amount + night 22-03 + shopping_net / misc_net

Combined Precision (%)	Combined Recall (%)	Combined F1 (%)	Combined FP
42.3%	30.0%	35.1%	3,947

Key Findings & Recommendations

What we found

- Fraud is 0.52% of transactions but 3.95% of dollar volume, so fraud tickets are much larger than normal ones.
- Hours 22-03 hold about 85% of fraud with about 23% of transactions.
- shopping_net, misc_net, shopping_pos and grocery_pos drive most of the fraud loss.

What the current rule does

- At $250: precision 13.3%, recall 74.5%, so most alerts are false alarms.
- At $825: precision 35.1%, recall 34.0%, so fewer false alarms but about two thirds of fraud is missed.
- No single amount threshold gives both high precision and high recall.

Recommendation

- Tested rule: amount + night window (22-03) + shopping_net/misc_net. At $250 it raises precision from 13.3% to 42.3% and cuts false alarms by about 92%, but recall falls from 74.5% to 30.0%.
- A layered approach could use the combined rule for automatic blocks and the amount-only rule for manual review.
- Choose the final threshold by weighing the cost of a false alarm against the cost of a missed fraud.

Limitations

- The data is simulated, so patterns may not match real-world fraud.
- The late-2020 fraud rate may be affected by data coverage.
- Findings describe the full dataset, not filtered views.
- The combined rule's conditions were chosen from this same dataset, so its performance may be overstated.